

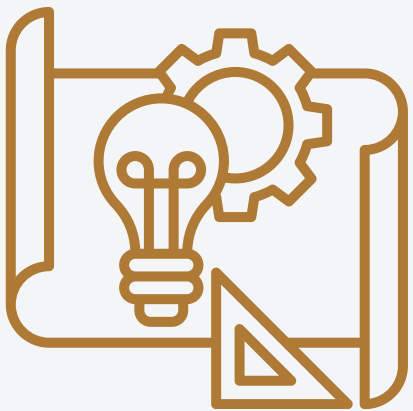
AI-POWERED PERSONAL HAJJ E-GUIDE

Students: Dania Alharbi 431201854 | Eman Alkhoudar 421202305
Farah Almutlaq 431201459 | Shahad Alkhattaf 431201570
Supervisor: Dr. Salim EL KHEDIRI
IT Department, College of Computer



◆ INTRODUCTION ◆

Millions of Muslims perform Hajj annually, yet existing digital solutions remain fragmented, connectivity dependent, and not pilgrim centric leaving individuals without personalized guidance or reliable offline support. Unlike current apps, **this project uniquely combines AI-powered ritual guidance, offline geofencing, and hybrid positioning into a single intelligent companion that functions without internet connectivity.**



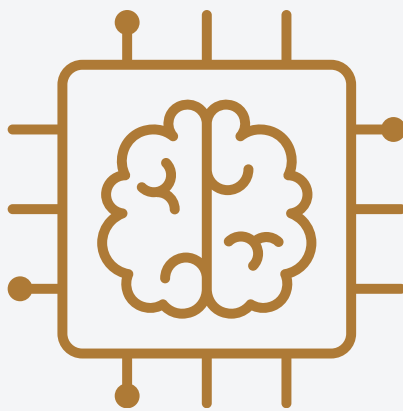
Architecture

Flutter → Python AI
backend → SQLite



Data

Mansak Al-Maysar
Arabic, English, Urdu



RAG + LLM

- User query is processed
- Retrieval using FAISS + BM25
- LLM generates a grounded answer based on retrieved context



Geofencing

Offline boundaries for Al-Haram, Mina, Arafat, and Muzdalifah trigger ritual instructions on zone entry

METHODOLOGY

◆ RESULTS ◆

~90% retrieval accuracy on real-world queries, with fully grounded responses, correct zone detection, and stable offline operation using lightweight embedding and language models.

◆ CONCLUSION ◆

The system delivers personalized, offline-capable Hajj guidance via RAG-based AI and geofencing, with future work targeting multilingual expansion and real-world Hajj testing.

